

# EEE 554: Exam 1 Study Guide

## Modules 1 and 2

Explanations, worked examples, and self-check questions

**How to use this guide.** Read the explanation, cover the worked answer, and attempt the problem yourself. Exam problems usually test whether you can choose the correct probability tool and explain why it applies, not only substitute values into a formula.

## Quick Formula Locator

### 1. Events, unions, intersections, and complements

$$P(A^c) = 1 - P(A)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$A \cap B = \emptyset \Rightarrow P(A \cup B) = P(A) + P(B)$$

### 2. Conditional probability

$$P(A|B) = P(A \cap B) / P(B)$$

$$P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$$

### 3. Bayes rule and total probability

$$P(A|B) = P(B|A)P(A) / P(B)$$

$$P(B) = \sum_i P(B|A_i)P(A_i)$$

$$P_e = P(Y=0|X=1)P(X=1) + P(Y=1|X=0)P(X=0)$$

### 4. Independence

$$P(A \cap B) = P(A)P(B)$$

$$P(A|B) = P(A)$$

### 5. Random variables and support

$$S_Y = \{y: y = g(x), x \in S_X\}$$

$$P(X=x) = 0 \text{ for a continuous random variable}$$

### 6. PMF, PDF, and CDF

$$F_X(x) = P(X \leq x)$$

$$P(a < X \leq b) = F_X(b) - F_X(a)$$

$$F_X(x) = \int_{-\infty}^x f_X(t) dt, \quad f_X(x) = dF_X(x)/dx$$

$$P(X = x_0) = F_X(x_0) - F_X(x_0^-)$$

## 7. Standard distributions

$$\text{Bernoulli: } E[X] = p, \quad \text{Var}(X) = p(1-p)$$

$$\text{Binomial: } P(X=k) = \binom{N}{k} p^k (1-p)^{N-k}$$

$$\text{Poisson: } P(X=k) = e^{-\lambda} \lambda^k / k!, \quad E[X] = \text{Var}(X) = \lambda$$

$$\text{Uniform}(a,b): f(x) = 1/(b-a), \quad E[X] = (a+b)/2, \quad \text{Var}(X) = (b-a)^2/12$$

$$\text{Exponential: } f(x) = \lambda e^{-\lambda x} u(x), \quad F(x) = 1 - e^{-\lambda x}, \quad E[X] = 1/\lambda$$

## 7b. Binomial $\rightarrow$ Poisson limit

$$M \rightarrow \infty, \quad p \rightarrow 0, \quad Mp = \lambda \text{ fixed} \Rightarrow \binom{N}{k} p^k (1-p)^{N-k} \rightarrow e^{-\lambda} \lambda^k / k!$$

## 8. Expectation and variance

$$E[X] = \sum x_i p_X(x_i) \quad \text{or} \quad E[X] = \int x f_X(x) dx$$

$$E[g(X)] = \sum g(x_i) p_X(x_i) \quad \text{or} \quad E[g(X)] = \int g(x) f_X(x) dx$$

$$\text{Var}(X) = E[X^2] - (E[X])^2, \quad E[aX+b] = aE[X] + b, \quad \text{Var}(aX+b) = a^2 \text{Var}(X)$$

## 8b. Chebyshev's inequality

$$\Pr(|X - \mu_X| > \gamma) \leq \sigma_X^2 / \gamma^2$$

## 9. Gaussian RVs and Q-function

$$f_X(x) = (1/\sqrt{2\pi\sigma^2}) e^{-(x-\mu)^2/(2\sigma^2)}$$

$$Z = (X - \mu)/\sigma, \quad P(X > x) = Q((x - \mu)/\sigma)$$

$$Q(x) = (1/\sqrt{2\pi}) \int_x^\infty e^{-t^2/2} dt$$

## 10–13. Transformations

$$F_Y(y) = P(g(X) \leq y)$$

$$f_Y(y) = f_X(g^{-1}(y)) |d[g^{-1}(y)]/dy|$$

$$Y = aX + b: f_Y(y) = (1/|a|) f_X((y-b)/a)$$

$$\text{Many-to-one: } f_Y(y) = \sum f_X(x_i) / |g'(x_i)|$$

$$\text{Clipping: } P(Y = x_{\max}) = 1 - F_X(x_{\max})$$

## Part 1: Probability Language

### 1. Events, unions, intersections, and complements

#### Equations to know

$$P(A^c) = 1 - P(A)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$A \cap B = \emptyset \Rightarrow P(A \cup B) = P(A) + P(B)$$

A probability problem starts with a sample space, the set of all possible outcomes. An event is a collection of outcomes. The notation tells you how events combine:

$$A \cup B: A \text{ or } B \text{ or both} \quad A \cap B: A \text{ and } B \quad A^c: \text{not } A$$

**Why it matters:** Before calculating, translate the words. “At least one” often means a union. “Both” means an intersection. “Not” means a complement.

**Worked example.** Suppose  $P(A) = 0.4$ ,  $P(B) = 0.5$ , and  $P(A \cap B) = 0.2$ . Find  $P(A \cup B)$ .

1. Add  $P(A)$  and  $P(B)$ .
2. The overlap  $A \cap B$  was counted twice, so subtract it once.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.4 + 0.5 - 0.2 = 0.7$$

**Self-check.** If  $P(A) = 0.35$ , find  $P(A^c)$ . **Answer:** 0.65.

### 2. Conditional probability

#### Equations to know

$$P(A|B) = P(A \cap B) / P(B)$$

$$P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$$

Conditional probability means the available sample space has changed because you are told that another event occurred. The vertical bar means “given.”

**Read the notation aloud:**

- $P(A|B)$  means “the probability of A given B.”
- $P(B|A)$  means “the probability of B given A.”
- $P(A \cap B)$  means “the probability that A and B both occur.”
- $P(A \cup B)$  means “the probability that A or B or both occur.”

**Important:** The event after the vertical bar is the information you already know happened. In  $P(A|B)$ , start inside event B, then ask what fraction of B also belongs to A. That is why  $P(B)$  is in the denominator.

**Word-to-math translation example.**

Let A = “the receiver gets a 1” and B = “the transmitter sent a 1.”

| Words                                                | Math notation | Meaning                                        |
|------------------------------------------------------|---------------|------------------------------------------------|
| Probability receiver gets 1                          | $P(A)$        | No condition is given.                         |
| Probability transmitter sent 1                       | $P(B)$        | No condition is given.                         |
| Probability receiver gets 1 given transmitter sent 1 | $P(A B)$      | Restrict attention to trials where B occurred. |
| Probability transmitter sent 1 given receiver gets 1 | $P(B A)$      | Restrict attention to trials where A occurred. |
| Probability transmitter sent 1 and receiver gets 1   | $P(A \cap B)$ | Both happen in the same trial.                 |

For the communication variables directly:

$P(Y=1|X=1)$  = “probability the receiver gets 1 given the transmitter sent 1”

$P(X=1|Y=1)$  = “probability the transmitter sent 1 given the receiver gets 1”

These are not generally equal because they answer different questions.

**Worked example.** A system sends a 1 with probability 0.4. It correctly receives a 1 given a transmitted 1 with probability 0.9. Find  $P(X=1, Y=1)$ .

1. The requested event is an intersection: transmitted 1 and received 1.
2. Use the multiplication rule.

$$P(X=1, Y=1) = P(Y=1|X=1)P(X=1) = 0.9(0.4) = 0.36$$

### 3. Bayes rule and total probability

#### Equations to know

$$P(A|B) = P(B|A)P(A)/P(B)$$

$$P(B) = \sum_i P(B|A_i)P(A_i)$$

$$P_e = P(Y=0|X=1)P(X=1) + P(Y=1|X=0)P(X=0)$$

Bayes rule reverses a conditional probability. In communications, you may know the channel behavior  $P(Y|X)$ , but want to infer what was transmitted after observing  $Y$ , which is  $P(X|Y)$ .

When  $P(B)$  is not known, find it by splitting  $B$  into mutually exclusive ways it can occur.

**Worked example.** A binary transmitter has  $P(X=1)=0.4$  and  $P(X=0)=0.6$ . The channel has  $P(Y=1|X=1)=0.9$  and  $P(Y=1|X=0)=0.1$ . Find  $P(X=1|Y=1)$ .

1. Find  $P(Y=1)$  using total probability.

$$P(Y=1) = 0.9(0.4) + 0.1(0.6) = 0.42$$

2. Apply Bayes rule.

$$P(X=1|Y=1) = 0.9(0.4)/0.42 = 0.857$$

Receiving a 1 makes it much more likely that a 1 was sent, but it is not certain because the channel can make errors.

**Self-check.** In the same system, find the probability of error. **Answer:**

$$P_e = P(Y=0|X=1)P(X=1) + P(Y=1|X=0)P(X=0) = 0.1(0.4) + 0.1(0.6) = 0.10$$

### 4. Independence

#### Equations to know

$$P(A \cap B) = P(A)P(B)$$

$$P(A|B) = P(A)$$

Events are independent when knowing one occurred does not change the probability of the other.

**Do not confuse independence with mutual exclusivity.** Mutually exclusive events cannot happen together. Independent events can happen together, but neither changes the chance of the other.

## Part 2: Random Variables and Distributions

### 5. What is a random variable?

#### Equations to know

$$S_Y = \{y: y=g(x), x \in S_X\}$$

$$P(X=x)=0 \quad \text{for a continuous random variable}$$

A random variable maps each experimental outcome to a number. It lets us use algebra, plots, and probability distributions to describe uncertain quantities such as number of errors, noise voltage, or waiting time.

**Discrete RV:** takes separated/countable values, like number of packet arrivals. **Continuous RV:** can take values on an interval, like noise voltage or transmission time. A continuous RV has zero probability of equaling one exact number.

### 6. PMF, PDF, and CDF

#### Equations to know

$$F_X(x) = P(X \leq x)$$

$$P(a < X \leq b) = F_X(b) - F_X(a)$$

$$F_X(x) = \int_{-\infty}^x f_X(t) dt, \quad f_X(x) = dF_X(x)/dx$$

$$P(X=x_0) = F_X(x_0) - F_X(x_0^-)$$

The PMF describes probability at each possible discrete value. The PDF describes how concentrated probability is around continuous values. The CDF works for every type of RV and gives probability up to a threshold.

$$\text{Discrete: } P(X=x_i) = p_X(x_i) \quad \text{Continuous: } P(a \leq X \leq b) = \int_a^b f_X(x) dx$$

**How to read a CDF:** It starts at 0, never decreases, and ends at 1. A jump means a point mass. A smooth section means continuous probability density.

**Worked example: CDF interval probability.** Suppose  $F_X(2)=0.75$  and  $F_X(1)=0.30$ . Find  $P(1 < X \leq 2)$ .

$$P(1 < X \leq 2) = F_X(2) - F_X(1) = 0.75 - 0.30 = 0.45$$

**Worked example: exponential PDF.** If  $X$  is exponential with rate  $\lambda=2$ , then  $f_X(x)=2e^{-2x}$ ,  $x \geq 0$ . Find  $P(0.5 \leq X \leq 1)$ .

1. Integrate the PDF, or subtract CDF values.

$$P(0.5 \leq X \leq 1) = \int_{0.5}^1 2e^{-2x} dx = e^{-1} - e^{-2} = 0.2325$$

**Self-check.** Why is  $P(X=2)=0$  for a continuous RV? **Answer:** A single point has zero area under a continuous PDF.

## 7. Standard distributions: recognize the story first

### Equations to know

$$\text{Bernoulli: } E[X]=p, \text{ Var}(X)=p(1-p)$$

$$\text{Binomial: } P(X=k) = \binom{N}{k} p^k (1-p)^{N-k}, \quad E[X]=Np, \text{ Var}(X)=Np(1-p)$$

$$\text{Poisson: } P(X=k) = e^{-\lambda} \lambda^k / k!, \quad E[X]=\text{Var}(X)=\lambda$$

$$\text{Uniform: } f(x)=1/(b-a), \quad E[X]=(a+b)/2, \quad \text{Var}(X)=(b-a)^2/12$$

$$\text{Exponential: } f(x)=\lambda e^{-\lambda x} u(x), \quad F(x)=1-e^{-\lambda x}, \quad E[X]=1/\lambda, \quad \text{Var}(X)=1/\lambda^2$$

**Bernoulli:** one trial, success/failure. **Binomial:** number of successes in  $N$  independent Bernoulli trials. **Poisson:** number of independent arrivals/events in an interval at a constant rate. **Uniform:** every value in an interval is equally likely. **Exponential:** waiting time until the next event at a constant rate. **Gaussian:** continuous variation from many small independent effects.

## 7b. Binomial $\rightarrow$ Poisson limit

### Equations to know

$$M \rightarrow \infty, \quad p \rightarrow 0, \quad Mp = \lambda \text{ fixed} \Rightarrow \binom{M}{k} p^k (1-p)^{M-k} \rightarrow e^{-\lambda} \lambda^k / k!$$

When the number of trials  $M$  is very large and the per-trial success probability  $p$  is very small, but their product  $\lambda=Mp$  stays a fixed, moderate number, the Binomial( $M,p$ ) PMF converges to the Poisson( $\lambda$ ) PMF. This is why Poisson is the standard model for "rare events across many opportunities": defective parts in a large batch, typos across many pages, packet arrivals in a network.

**Rule of thumb** for when the approximation is good:  $M$  large (say  $\geq 50$ ),  $p$  small (say  $\leq 0.05$ ), and  $\lambda=Mp$  a moderate, fixed value — not growing without bound.

**Worked example.** A batch of  $M=1000$  components each has an independent defect probability  $p=0.004$ . Exactly,  $X \sim \text{Binomial}(1000, 0.004)$ . Approximate the distribution of  $X$ .

1. Compute  $\lambda = Mp$ .

$$\lambda = 1000(0.004) = 4$$

2. Replace the binomial PMF with the Poisson(4) PMF.

$$P(X=k) \approx e^{-4} 4^k / k!$$

**Self-check.** If  $M=500$  and  $p=0.008$ , what  $\lambda$  would you use for the Poisson approximation? **Answer:**  $\lambda = Mp = 500(0.008) = 4$ .

## Part 3: Mean, Variance, and Gaussian Probability

### 8. Expectation and variance

#### Equations to know

$$E[X] = \sum x_i p_X(x_i) \quad \text{or} \quad E[X] = \int x f_X(x) dx$$

$$E[g(X)] = \sum g(x_i) p_X(x_i) \quad \text{or} \quad E[g(X)] = \int g(x) f_X(x) dx$$

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

$$E[aX+b] = aE[X] + b, \quad \text{Var}(aX+b) = a^2 \text{Var}(X)$$

The mean is the balance point or long-run average. Variance measures spread around the mean.

**Worked example.** Let  $X$  be Bernoulli with  $p=0.3$ . Then  $X$  is 1 with probability 0.3 and 0 with probability 0.7.

$$E[X] = 1(0.3) + 0(0.7) = 0.3$$

$$E[X^2] = 1^2(0.3) + 0^2(0.7) = 0.3$$

$$\text{Var}(X) = E[X^2] - E[X]^2 = 0.3 - 0.3^2 = 0.21$$

**Transformation rules:** Adding a constant shifts values but does not change spread. Multiplying by a constant stretches the spread, so variance uses the square.

### 8b. Chebyshev's inequality

#### Equations to know

$$\Pr(|X - \mu_X| > \gamma) \leq \sigma_X^2 / \gamma^2$$

Chebyshev's inequality bounds how much probability can lie far from the mean using only the mean and variance — no knowledge of the full distribution is required. It applies to *any* random variable with finite variance, discrete or continuous.

**Why it matters:** It's a bound, not an exact value, and it is often loose (a Gaussian assumption, when justified, gives a much tighter tail estimate — see Q-function below). Use Chebyshev when you're only given a mean and variance and no distributional form.

**Worked example.** A measurement  $X$  has  $\mu=10$ ,  $\sigma^2=4$  (distribution unknown). Bound  $\Pr(|X-10|>5)$ .

$$\Pr(|X-10|>5) \leq \sigma^2 / \gamma^2 = 4 / 5^2 = 4/25 = 0.16$$

The probability of straying more than 5 from the mean is at most 16%, regardless of X's actual distribution.

**Self-check.** If  $\sigma^2=9$ , find  $\gamma$  such that Chebyshev guarantees  $\Pr(|X-\mu|>\gamma)\leq 0.09$ . **Answer:**  $\gamma^2=\sigma^2/0.09=9/0.09=100 \Rightarrow \gamma=10$ .

## 9. Gaussian random variables and the Q-function

### Equations to know

$$f_X(x) = (1/\sqrt{2\pi\sigma^2})e^{-(x-\mu)^2/(2\sigma^2)}$$

$$Z = (X-\mu)/\sigma, \quad Z \sim N(0,1)$$

$$P(X > x) = Q((x-\mu)/\sigma)$$

$$Q(x) = (1/\sqrt{2\pi}) \int_x^\infty e^{-t^2/2} dt$$

If  $X \sim N(\mu, \sigma^2)$ , convert it to a standard normal variable before using a table or Q-function.

**Worked example.** Let  $X \sim N(3, 1)$ . Find  $P(X > 8)$ .

1. Standardize the threshold.

$$z = (8-3)/1 = 5$$

2. Use the right-tail function.

$$P(X > 8) = Q(5) \approx 2.87 \times 10^{-7}$$

This is a rare event because 8 lies five standard deviations above the mean.

## Part 4: Transformations of Random Variables

### 10. The main question: what values can the output take?

#### Equations to know

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y)$$

$$S_Y = \{y: y = g(x), x \in S_X\}$$

For  $Y=g(X)$ , do not begin with a formula. First map the support of  $X$  through  $g$ . Then decide whether each output value comes from one input or several inputs.

#### Reliable procedure

1. Find the support of  $X$ .
2. Map it through  $g$  to find the support of  $Y$ .
3. Find every inverse value of  $x$  that gives the desired  $y$ .
4. Use the inverse/Jacobian formula if the mapping is one-to-one, or add contributions from each branch if it is many-to-one.
5. Check whether an interval of  $X$  collapsed into a single  $Y$  value. If so, include a point mass.

### 11. One-to-one affine transformation

#### Equations to know

$$f_Y(y) = f_X(g^{-1}(y)) |d[g^{-1}(y)]/dy|$$

$$Y = aX + b \Rightarrow f_Y(y) = (1/|a|) f_X((y-b)/a)$$

**Worked example.**  $X \sim U(0,2)$ , and  $Y=3X+1$ .

1. Map endpoints: 0 maps to 1 and 2 maps to 7. Therefore  $1 \leq Y \leq 7$ .
2. The original uniform density is  $1/2$ .
3. The scale factor is  $1/|3|$ .

$$f_Y(y) = (1/3)(1/2) = 1/6, \quad 1 \leq y \leq 7$$

So  $Y \sim U(1,7)$ .

### 12. Many-to-one transformation

#### Equations to know

$$f_Y(y) = \sum_{\text{valid roots } x_i} f_X(x_i) / |g'(x_i)|$$

$$Y = |X|: f_Y(y) = f_X(y) + f_X(-y), \quad y \geq 0$$

For  $Y = X^2$  or  $Y = |X|$ , more than one input can produce the same output. You must include every valid input branch.

**Worked example.** If  $X \sim N(0, 1)$  and  $Y = X^2$ , then  $y > 0$  has two roots:  $x = \sqrt{y}$  and  $x = -\sqrt{y}$ . Both have derivative magnitude  $2/\sqrt{y}$ .

$$f_Y(y) = e^{-y/2} / \sqrt{(2\pi y)}, \quad y > 0$$

Because a square cannot be negative,  $f_Y(y) = 0$  for  $y \leq 0$ .

### 13. Clipping creates a mixed RV

#### Equations to know

$$P(Y = x_{\max}) = P(X \geq x_{\max}) = 1 - F_X(x_{\max})$$

$$P(Y = y_0) = F_Y(y_0) - F_Y(y_0^-)$$

If  $Y = \min(X, x_{\max})$ , every input value above  $x_{\max}$  becomes the same output value  $x_{\max}$ . The output has a continuous portion below  $x_{\max}$  plus a discrete point mass at  $x_{\max}$ .

**Final self-check.** Why can you not use only a smooth PDF for a clipped RV? **Answer:** The clipped values create nonzero probability at one exact output, which is a point mass.

## Final Exam Practice Routine

1. Do the Module 1 binary communication example until Bayes, total probability, and error probability feel automatic.
2. Practice converting PMF/PDF/CDF in both directions and reading interval probabilities from a CDF.
3. For every common distribution, say the physical story, support, mean, and variance — including when Binomial collapses to Poisson ( $M$  large,  $p$  small,  $Mp = \lambda$  fixed).
4. Practice three transformations: affine, absolute value/square, and clipping.
5. Know Chebyshev's inequality as a distribution-free fallback whenever a problem gives you only a mean and variance.
6. When stuck, write the definition first: conditional probability, CDF, expectation, variance, or transformed CDF.